

# DAVID OPREA

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## Education

**Purdue University** West Lafayette, IN  
*B.S. in Computer Science* Expected May 2028  
GPA: 3.9/4.0 | **Relevant Coursework:** Multivariable Calculus, CS Tools (Linux/Git/CLI)

## Skills

- **Languages:** Python, Java, C++ , JavaScript, Julia, HTML/CSS
- **Frameworks & Tools:** PyTorch, NumPy, Pandas, Matplotlib, React, Electron, Node.js, Flask, Git, GitHub
- **Topics:** Machine Learning, Quant Finance, Full Stack, Data Visualization, Algorithmic Optimization

## Technical Experience

- **MascotGO** — *Software Engineering Intern (Student in Residence)* Dec 2025 – Present
  - Leading the research and development of a comprehensive sports identity database covering **1,100+ NCAA schools** for integration into MascotGO's AI knowledge graph for future lookup features.
  - Developing an AI-driven social insight pipeline using **multi-agent workflows** and **Playwright** to extract and validate authentic student sentiment from online platforms at scale.
  - Implementing **data validation tests** and **cleaning pipelines** to ensure accuracy of static identity data, establishing quality standards for college matching recommendations.
- **Boiler Quant** — *Quantitative Analyst* Sept 2025 – Present
  - Engineered a **liquidity rewards system** and a user deposit infrastructure through integration with **Stripe's SDKs** for a social stock exchange, designing incentivization mechanics for Market Makers.
  - Implemented a full-featured order book using **TypeScript** and a **RESTful API** architecture to perform operations, establishing core exchange market mechanics and transaction logic.
- **Purdue Stack** — *Full Stack Developer* Sept 2025 – Dec 2025
  - Finalized a standalone desktop GUI for seismologists using **Electron**, **React**, and **Flask** enabling high-fidelity data analysis of seismic events, stations, and waveforms; currently utilized by Purdue faculty.
  - Integrated **Plotly** and **D3.js** for interactive waveform plots and spectrograms, supporting imports of JSON, CSV, and miniSEED files and also exporting plot and waveform visuals for scientific reporting.

## Research & Projects

- **ML@Purdue** — *Machine Learning Researcher (VGC-RL Team)* Sept 2025 – Present
  - Architecting a Deep RL agent for competitive Pokémon VGC in **PyTorch**, utilizing an off policy-driven neural network and tinyBERT model to optimize complex move-selection in high-dimensional state spaces.
  - Optimized agent performance by creating a high-throughput policy model and behavior cloning implementation, improving the amount of information for states stored, speed, and training convergence.
- **ReclarifAI** — *AI-Powered Learning App* Hackathon Project
  - Engineered a full-stack **Flask/React** app using the **OpenAI API** to generate semantic-based definitions in real-time, improving active recall over traditional flashcard software.

## Awards & Honors

- **USACO Platinum Division** (Top 200 U.S. High School Competitors)
- **National Cyber Scholarship Finalist** (Top 1000 U.S. High School Students)
- **Science Olympiad:** 40+ medals, Redox probe with <5% error; 1st Place — Princeton Invitational.